

ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY & SCIENCES

Autonomous Institution Affiliated to Andhra University | Approved by AICTE

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Course: Full Stack Web Development Lab

Lab Record Week: #04

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1. TITLE OF THE EXPERIMENT

Development of a Dynamic Web Application with Live Clock, Task Manager & Themes

2. AIM

To build an interactive dynamic webpage combining real-time digital clock updates (setInterval), interactive greeting message generation, dynamic task list management, and live background theme switching using JavaScript DOM manipulation.

3. PROCEDURE / ALGORITHM

Step 1: Construct HTML layout with cards for Welcome greeting, Live Clock, Task Manager, and Colour Theme switcher.

Step 2: Implement welcome() function to read input name and dynamically display greeting message.

Step 3: Implement clock() function using new Date().toLocaleTimeString() and run every second via setInterval(clock, 1000).

Step 4: Implement addTask() function to append new items to <ul id='list'>.

Step 5: Implement blue(), green(), and dark() theme functions modifying document.body.style.background with linear gradients.

4. TEST CASES / VERIFICATION

Test Case	Description / Action	Expected Output	Status
TC-01	Welcome Greeting	Displays 'Welcome <Name>' upon clicking Submit	Passed
TC-02	Live Digital Clock	Clock updates seconds in real-time every 1000ms	Passed
TC-03	Task Manager Addition	Adds new task to list when task field is non-empty	Passed
TC-04	Live Theme Switching	Switches background gradient between Blue, Green, and Dark themes	Passed

5. LEARNING OUTCOME / RESULT

The dynamic multi-feature webpage was successfully created and verified, demonstrating real-time DOM timers, dynamic element insertion, and live style manipulation.

6. SOURCE CODE

File: [index.html](#)

```
<!DOCTYPE html>
<html>
<head>
<title>Dynamic Webpage</title>
<style>
*{margin:0;padding:0;box-sizing:border-box}
body{
  font-family:Arial;
  background:linear-gradient(135deg,#4facfe,#00f2fe);
  min-height:100vh;
}
header{
  background:#222;color:white;text-align:center;padding:20px;
}
.container{
  width:95%;max-width:1200px;margin:20px auto;
}
.card{
  background:white;padding:35px;min-height:200px;
  margin-bottom:20px;border-radius:15px;
  box-shadow:0 5px 15px #555;
}
input{
  padding:12px;width:65%;margin:15px 0;
}
button{
  padding:10px 15px;border:0;border-radius:5px;
  background:#007bff;color:white;cursor:pointer;
}
button:hover{background:#0056b3}
#clock{font-size:35px;margin-top:25px}
li{margin:8px;padding:8px;background:#eee}
</style>
</head>

<body>

<header>
<h1>Dynamic Webpage</h1>
<p>Using HTML, CSS and JavaScript</p>
</header>

<div class="container">

<div class="card">
<h2>Welcome</h2>
<input id="name" placeholder="Enter your name">
<button onclick="welcome()">Submit</button>
<p id="msg"></p>
</div>

<div class="card">
<h2>Live Clock</h2>
<p id="clock"></p>
</div>

<div class="card">
<h2>Task Manager</h2>
<input id="task" placeholder="Enter task">
<button onclick="addTask()">Add</button>
<ul id="list"></ul>
</div>

<div class="card">
<h2>Colour Theme</h2>
<button onclick="blue()">Blue</button>
</div>
</div>
```

```
<button onclick="green()">Green</button>
<button onclick="dark()">Dark</button>
</div>

</div>

<script>
function welcome(){
  let n=document.getElementById("name").value;
  document.getElementById("msg").innerHTML="Welcome "+n;
}

function clock(){
  document.getElementById("clock").innerHTML=
  new Date().toLocaleTimeString();
}
setInterval(clock,1000);clock();

function addTask(){
  let t=document.getElementById("task").value;
  if(t!=""){
    let li=document.createElement("li");
    li.innerHTML=t;
    document.getElementById("list").appendChild(li);
  }
}

function blue(){
  document.body.style.background=
  "linear-gradient(135deg,#4facfe,#00f2fe)";
}
function green(){
  document.body.style.background=
  "linear-gradient(135deg,#43e97b,#38f9d7)";
}
function dark(){
  document.body.style.background=
  "linear-gradient(135deg,#232526,#414345)";
}
</script>

</body>
</html>
```

7. OUTPUT SCREENSHOTS

Output Screenshot: [Screenshot 2026-08-25 061830.png](#)

Dynamic Webpage

Using HTML, CSS and JavaScript

Welcome

Submit

Welcome MANOJ

Live Clock

6:18:20 am

Task Manager

Add

- fs
- cpp

Colour Theme

Blue

Green

Dark